Janssen effect

The Janssen effect is a characteristic feature of granular materials. When granules are poured into a container, a network of contacts between particles leads to the formation of force chains (see Fig.) that redistribute the stress of the granular column towards the side walls.

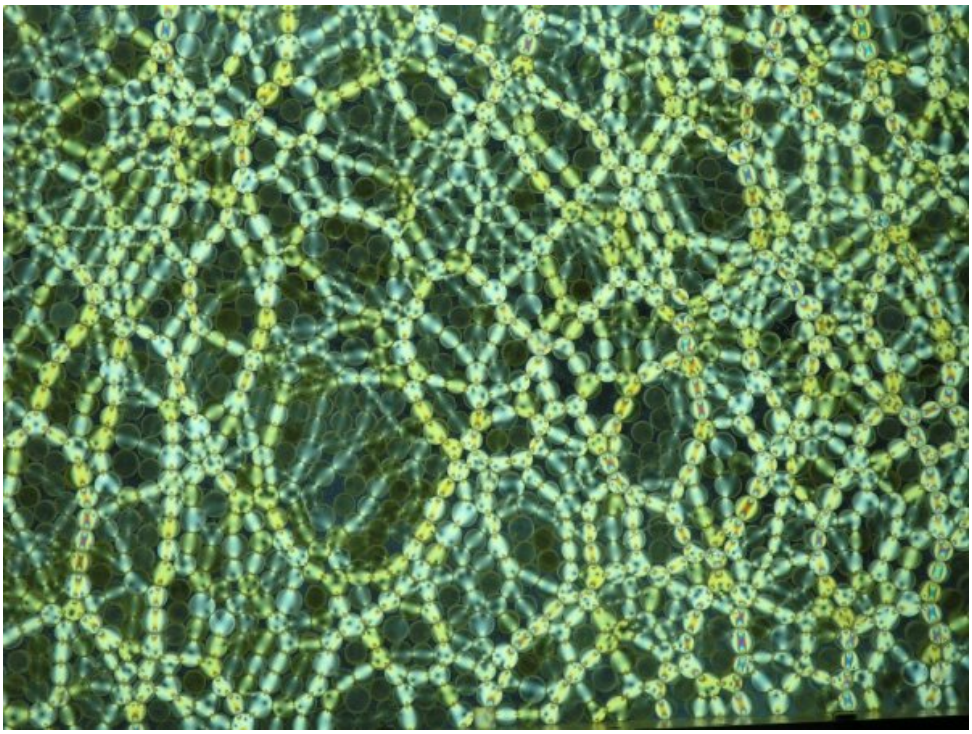


Fig. 1: Force chains

This ability of granular materials to self-organize stress distribution through force chains distinguishes them from conventional liquids and solids, making their study important for various fields of science and engineering. This is particularly important in industry, where large quantities of raw materials are typically stored as granules in silos (storage facilities for granular material).

This problem is devoted to investigating this effect by examining the behavior of spherical granules in vessels of various shapes.

Throughout the problem, disregard the contribution of atmospheric pressure. All pressures refer to gauge pressure relative to atmospheric pressure.

The following integrals may be useful:

$$\int \frac{dx}{a+x} = \ln|a+x| + C$$

$$\int x^n dx = \frac{1}{n+1}x^{n+1} + C, \quad n \neq -1$$

$$\int e^x dx = e^x + C$$

Part A. Hydrostatics (0.5 points)

A cylindrical vessel with vertical walls and a bottom area S (see Fig.) is completely filled with water of total mass m. The density of the water is ρ.

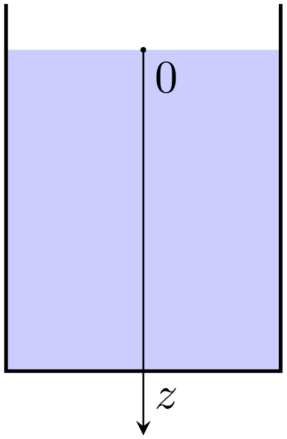


Fig. 2: Cylindrical vessel with water

A.1	Determine the pressure p_1 at the bottom of the vessel. Express your answer in terms of m, g, and S.	0.2pt

A.2	Determine the pressure $p(z)$ on the vessel walls as a function of depth z. Express the answer in terms of ρ, g, and z.	0.3pt

Part B. Janssen effect in a cylindrical vessel with vertical walls (2.9 points)

This part investigates how the pressure exerted by the granules on the bottom of a vertical-walled cylindrical vessel depends on the mass of the granules.

For parts **B**, **C** and **D** of the problem, work within the following model:

- ρ is the **bulk** density of the material (average mass of granules per unit volume of the vessel).
- Before the experiments, the vessel is shaken, ensuring a completely random arrangement of the granules. There is a very large number of granules, and the free surface is horizontal.
- z is the depth measured from the free surface ($z = 0$ at the averaged granule-air interface, with the z-axis directed vertically downward).
- Assume that the granular pressure against the walls is directly proportional to the vertical pressure in the vessel: $\sigma(z) = KP(z)$. Here, K is the constant coefficient. Throughout the remainder of the problem, understand the pressure of the granules as the averaged normal force acting per unit surface area. For example, for the pressure on the vessel walls:

$$\sigma(z) = \left\langle \frac{dF_{\text{hor}}}{dS}(z) \right\rangle .$$

- Assume this quantity is independent of the radial direction. Also, assume the vertical pressure is independent of the radial direction and the distance from the vessel axis.

- μ is the friction coefficient of the granular material against the vessel wall. Assume that the friction force acting on the granules is vertical and maximal in magnitude. Neglect the phenomenon of stagnation; that is, assume that the maximum possible value of the friction force is $dF = \mu\sigma dS$. In parts **B** and **C**, assume that friction minimizes the pressure on the bottom of the vessel, i.e., the friction forces acting on the granules are directed vertically upwards.
- R is the inner radius of the cylindrical vessel.
- $g = 9.8\ m/s^2$ is the acceleration due to gravity.

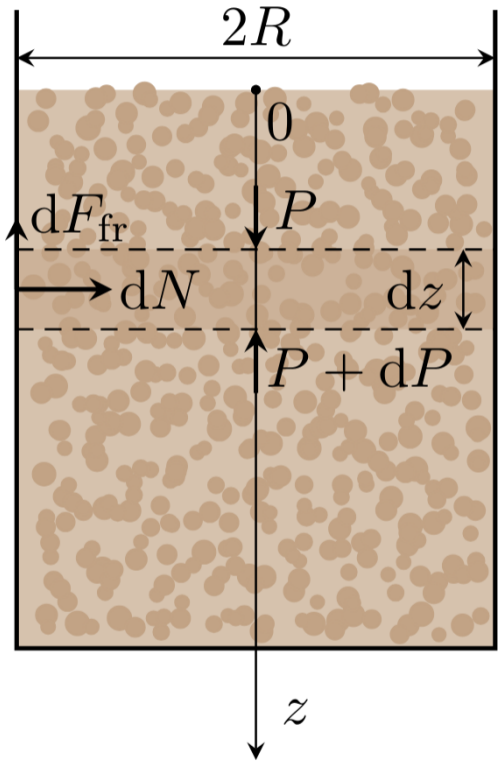


Fig. 3: Cylindrical vessel with granules

Consider a thin layer of granular material at depth z with thickness dz. The radius of this layer is equal to the inner radius of the vessel.

B.1 Write down the equilibrium condition for this layer. The equation may include the pressures $P(z)$ and $P(z+dz)$, dz, the average pressure on the vessel walls $\sigma(z)$, the mass of the layer dm, as well as μ, g, and R. 0.5pt

B.2 Show that the bottom pressure P as a function of the fill height h of the granular column can be written as: 1.2pt

$$P = A + Be^{-h/\lambda}.$$

Determine the parameters A, B, and λ. Express your answers in terms of the bulk density ρ, as well as g, R, K, and μ.

The graph below shows the dependence of the granular pressure **on the bottom** on the granules level in the vessel, $P_0(H)$.

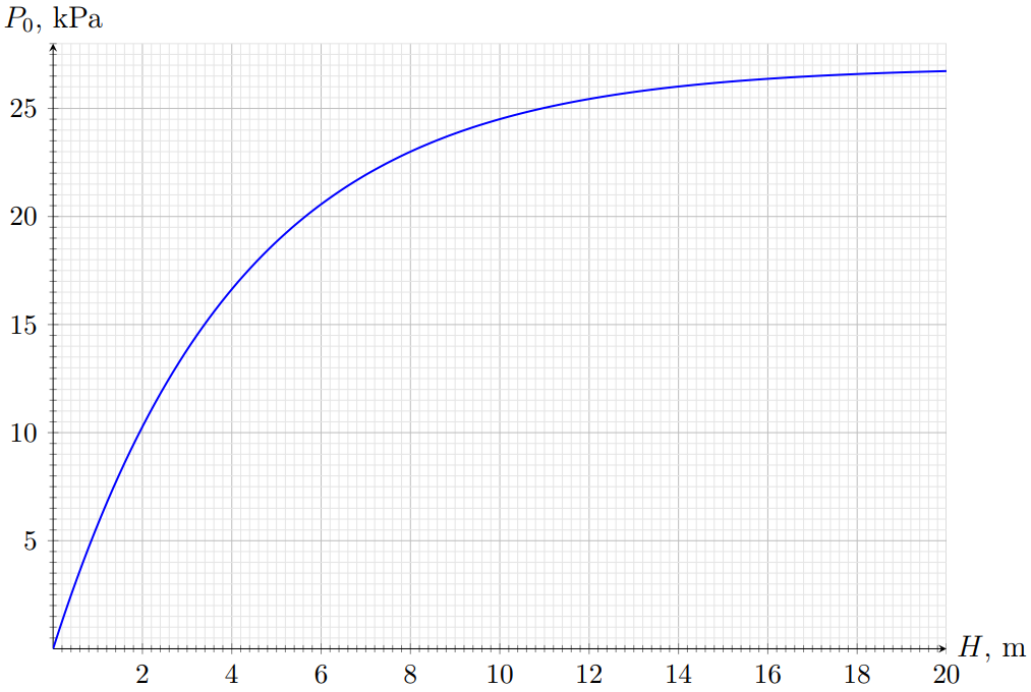


Fig. 4: Graph of the $P(h)$ dependence

B.3	Determine the bulk density of the granules ρ and the coefficient K. Give the answers to two significant figures. For this part, assume the numerical value of the vessel radius is $R = 50\ \mathrm{cm}$ and the coefficient of friction is $\mu = 0.30$.	1.2pt

Part C. Conical vessel (2.3 points)

In this part of the problem, we investigate the pressure distribution of granules in a conical vessel, oriented with the bottom facing upward and **completely** filled with granules. This calculation is crucial, for example, for designing grain storage silos (see the figure below).



Fig. 5: Conical silo

Assume the base radius R and the cone's half-angle $\alpha \ll 1$ are known.

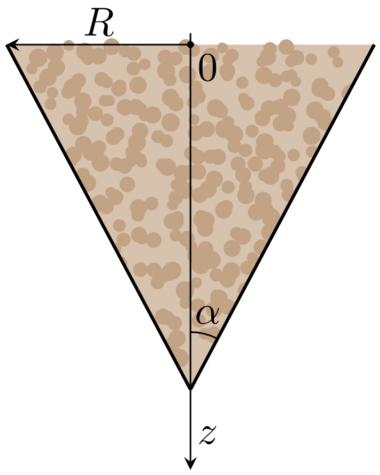


Fig. 6: Granules in a conical vessel

C.1	In the specified approximation, obtain the dependence of the vessel's cross-sectional radius on height $r(z)$. Express the answer in terms of R, α, and z.	0.1pt

Let us again consider a thin layer of granular material analogous to the layer in **B1**.

C.2	Write down the equilibrium condition for the layer. The equation may include the pressures $P(z)$ and $P(z+dz)$, dz, the average pressure on the vessel walls $\sigma(z)$, the mass of the layer dm, as well as μ, g, $r(z)$, $r(z+dz)$, α, and R.	0.6pt

For the remainder of this part, assume that $\alpha \ll \mu$. Note that $K\mu$ and α may be of the same order of magnitude. Let us introduce the notation:

$$z_{\text{max}} = \frac{R}{\alpha}.$$

Assume henceforth that the dependence of the vertical pressure P inside the vessel on the coordinate z is given by:

$$P(z) = A\left(1 - \frac{z}{z_{\text{max}}} - \gamma\left(1 - \frac{z}{z_{\text{max}}}\right)^{\beta}\right),$$

where A, γ, and β are constants. The system parameters are chosen such that $A, \beta > 0$.

C.3	Determine the parameters A, γ, and β. Express the answer in terms of the bulk density of the granules ρ, g, z_{max}, K, μ, and α.	0.8pt

C.4	Sketch a qualitative graph of $P(z)$ and determine the ratio of the maximum vertical pressure P_{max} to $\rho g z_{\text{max}}$ for $\alpha = 0.2K\mu$. Provide the answer to two significant figures.	0.8pt

Part D. What if the vessel is lifted? (1.5 points)

In this part of the problem, we again consider a vertical cylindrical massless vessel of radius R.

A layer of granular material of height H is poured into it. A rod of radius a is vertically inserted into the center of the container, touching the bottom. The bulk density of the granules is ρ, the coefficient of friction of the material against the container wall is μ_1, and the coefficient of friction of the material against the rod is μ_2. The rod begins to be slowly pulled upward at a constant low speed. Due to friction, the vessel with the granules begins to rise slowly. Throughout this motion, the granules remain stationary **relative to the vessel**.

The coefficient K is still constant both for the interaction of granules with the vessel walls and for the interaction of granules with the rod.

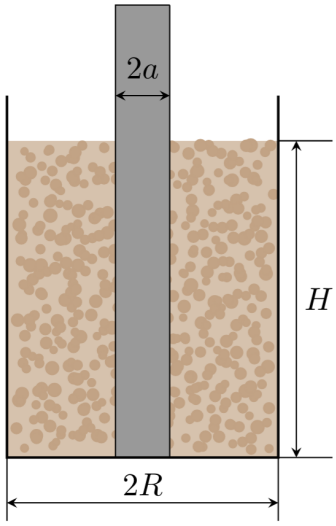


Fig. 7: Rod in a vessel with granules

Let us again consider a thin layer of granular material analogous to the layer in **B1**. Now, this layer is in contact with both the vessel and the rod.

D.1	Write down the equilibrium condition for the layer. The equation may include the pressures $P(z)$ and $P(z+dz)$, dz, the average pressure on the rod and vessel walls $\sigma(z)$, the mass of the layer dm, as well as μ_1, μ_2, g, a, and R.	0.4pt

D.2	Show that the bottom pressure P as a function of the fill height h of the granular column can be written as: $$P = A + Be^{-h/\lambda}.$$ Determine the parameters A, B, and λ. Express the answers in terms of the bulk density of the granules ρ, as well as g, a, R, μ_1, μ_2, and K.	0.3pt

D.3 Determine the minimum level H of the granules relative to the bottom at which the cylinder can be lifted as described above. Assume that $H \ll \lambda$. Express the answer in terms of a, R, μ_2, and K. 0.8pt
Hint: You may use the following approximation:

$$e^x \approx 1 + x + \frac{x^2}{2}, \quad x \ll 1$$

Part E. Fluid outflow through the wall (1.8 points)

An **incompressible** fluid characterized by the complete absence of internal friction (viscosity) and a constant density ρ is termed an **ideal** fluid. Hereafter, we shall focus our analysis on such an ideal fluid.

Consider a stream tube within a steady flow of an ideal fluid, bounded by small cross-sections S_1 and S_2, through which the fluid flows from left to right. Assume that the fluid pressure and velocity are uniform across any cross-section within this stream tube that is perpendicular to the fluid velocity.

Let the flow velocity v_1, pressure p_1, and height h_1 of this cross-section be given at S_1. Similarly, at cross-section S_2, the flow velocity v_2, pressure p_2, and height h_2 are specified. The height h is measured from a fixed reference level.

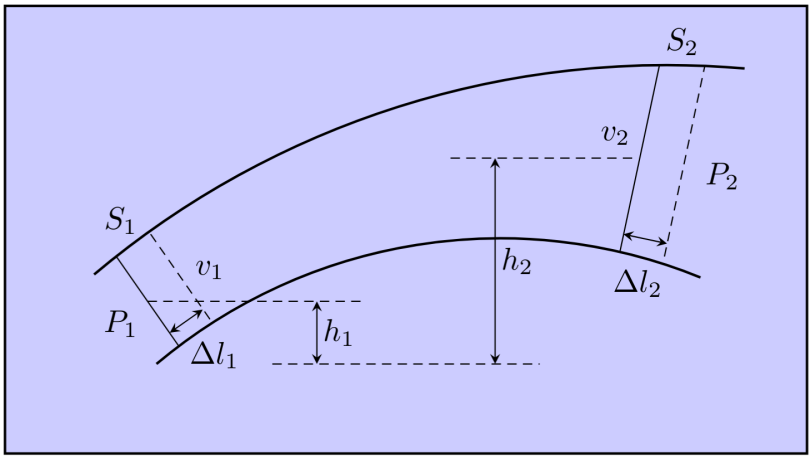


Fig. 8: Ideal fluid flow

Over a small time interval, a volume of fluid moves along the stream tube, with cross-section S_1 moving to position $S_1' \approx S_1$ over a distance Δl_1, and cross-section S_2 moving to position $S_2' \approx S_2$ over a distance Δl_2.

E.1 Derive an equation relating S_1, S_2, Δl_1, and Δl_2. 0.2pt

E.2 Determine the total energy E (the sum of kinetic and potential energies) of the selected fluid volume $S_1 \Delta l_1$. Express the answer in terms of ρ, S_1, Δl_1, v_1, g, and h_1. Take the potential energy at $h = 0$ as the zero reference level. 0.1pt

E.3	Determine the work A done by pressure forces on the fluid bounded by the chosen stream tube during time $\Delta t = \frac{\Delta l_1}{v_1}$. Express the answer in terms of p_1, p_2, S_1, and Δl_1.	0.3pt

E.4	Using the law of conservation of energy, express the pressure p_2 in terms of p_1, ρ, g, h_1, h_2, v_1 and v_2. The resulting equation is known as the **Bernoulli equation**.	0.3pt

Consider a cylindrical vessel with vertical walls, similar to the vessel in Part **B**. Its inner radius is R, and an hole with diameter $D \ll R$ is made in its wall. The vessel is filled with an ideal fluid of density ρ, and its height relative to the hole is $h \gg D$.

E.5	Determine the fluid outflow velocity $v(h)$ as a function of the height above the hole. Express the answer in terms of g and h.	0.3pt

At the initial time, let the fluid level above the hole be H.

E.6	Determine the time τ required for all the fluid initially above the hole to drain through it. Express the answer in terms of g, H, R, and D.	0.6pt

Part F. "Flow" of granules through a wall (1 point)

In this part of the problem, we examine the “flow” of granules from a cylindrical vessel, focusing on its characteristics and differences from the flow of an ideal fluid. To do this, we again consider a cylindrical vessel with vertical walls filled with granules. Assume each granule is a sphere of diameter d. A circular hole of diameter $D \gg d$ is made in the cylindrical vessel. The condition $D \gg d$ implies that the effective diameter of the opening does not decrease due to arching (accumulation of granules near it). Above the opening, there is a layer of granules of height $H \gg \lambda$, where λ is the parameter found in part **B2**. The vessel wall thickness is ω. Additionally, assume that $D \ll R$, where R is the internal radius of the vessel.

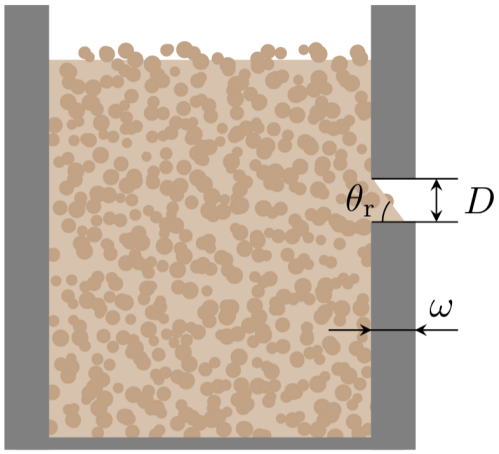


Fig. 9: Equilibrium of granules in a vessel with thick walls

F.1	Due to internal friction between the granules, the maximum angle at which the free surface of the granular material can deviate from the horizontal is θ_r. What is the maximum wall thickness ω_{max} for which the flow of granules from the vessel is possible? Express the answer in terms of D and θ_r.	0.2pt

Unlike an ideal fluid, significant energy losses occur in granular materials due to interparticle friction. Assume that the kinetic energy of the exiting particles is gained only at characteristic distances of the order of D from the hole, and the characteristic acceleration of the granules during acceleration is of the order of g.

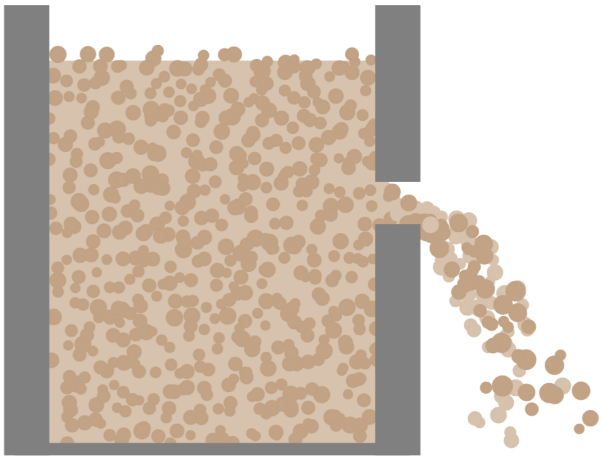


Fig. 10: Granule discharge through an orifice

Therefore, the mass flow rate of granules exiting through the hole is expressed as:

$$\frac{dm}{dt}(H, D) = AH^q D^n,$$

where A is a dimensional constant, and $H \gg D$ is the height of the layer of granules above the hole.

F.2	Determine q and n.	0.8pt

Air shower

Part A. Electromagnetic shower (2.0 points)

Bremsstrahlung

When passing through matter, charged particles interact with the atoms and molecules of the material and, as a result, lose their kinetic energy. There are various mechanisms of energy loss: ionization of matter, bremsstrahlung, production of electron-positron pairs, and others. The fraction of energy lost through each of these mechanisms depends on the parameters of the particle and its motion — its kinetic energy, the particle mass, and its charge. In this problem, we will consider ultrarelativistic ($\beta \simeq 1,\ E_{\text{kin}} \gg m_e c^2$) electrons (as well as *positrons* — positively charged electrons), for which the main mechanism of energy loss is bremsstrahlung.

In shower physics, the concept of **radiation path length** is introduced. It depends not only on the distance traveled, but also on the density of the medium in which the motion takes place. In fact, the radiation path length indicates the number of "layers" of matter passed through by the particle. The radiation path length of a particle in matter can be calculated using the formula

$$x = \int_{z_1}^{z_2} \rho(z) dz, \tag{1}$$

where z_1 and z_2 are the geometric coordinates of the beginning and end of the particle's motion (in meters), and $\rho(z)$ is the density of the medium.

An ultrarelativistic electron e^- (or positron e^+) moving in air can spontaneously emit a photon γ, which carries away part of its energy. This process is probabilistic and follows an exponential distribution in length. This means that if an electron starts moving along the x-axis from the point $x_0 = 0$, then the probability density that a photon will be emitted at the point with coordinate x is

$$f(x) = \frac{1}{X_0} e^{-x/X_0}, \tag{2}$$

where X_0 is the radiation length of the material. It is a constant and does not depend on the parameters of the particle's motion or on the density of the medium. In shower physics, the distance x is measured not in the usual units for us (meters), but in radiation lengths:

$$[x] = [X_0] = [\text{g/cm}^2]. \tag{3}$$

For air, the value is $X_0 = 36.2\ \text{g/cm}^2$.

High-energy photons passing through matter also interact with it. The interaction mechanisms include the photoelectric effect, Compton scattering, and electron-positron pair production. At high energies, the dominant energy-loss mechanism is electron-positron pair production. A photon interacts with the electric field of an atom and produces one electron and one positron. This process is also probabilistic and follows the same exponential distribution with the same radiation length X_0 as for electrons.

Heitler Model

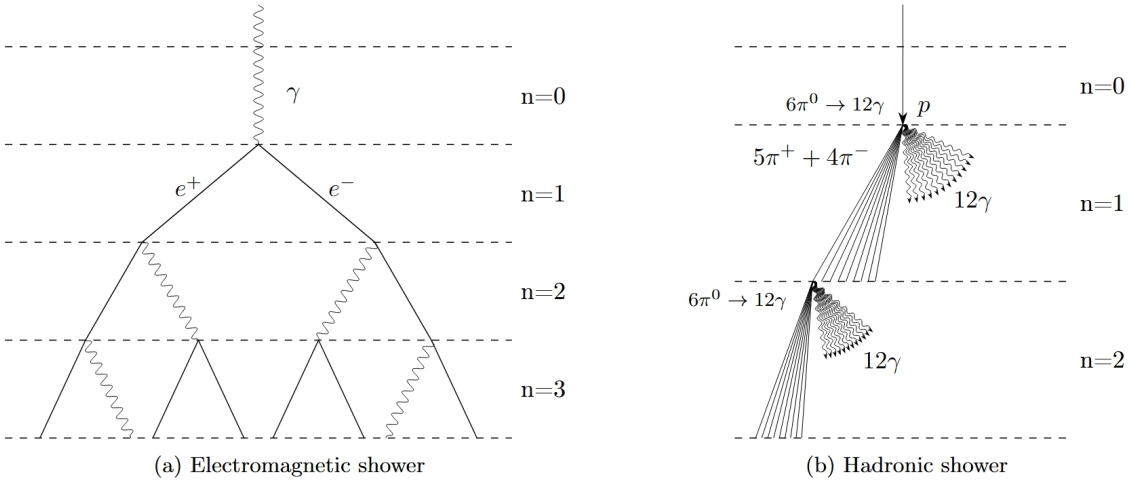
In this problem, we will work within the simplified Heitler model. In this model, bremsstrahlung is not a probabilistic process but a deterministic one. An electron moving through matter emits one photon every time it travels a certain radiation path length d_e, which we will call the **splitting length**. During this emission, the total energy of the electron is divided equally between the electron and the emitted

photon. The electron continues its motion and emits one photon every splitting length d_e until its energy falls below the critical energy E_e^{crit}, below which ionization losses in air begin to dominate and the development of the electromagnetic shower ceases. For an electron in air, $E_e^{\text{crit}} = 88\ \text{MeV} = 88 \cdot 10^6\ \text{eV}$. Accurate calculations show that the splitting length d_e is related to the radiation length X_0 of the material by

$$d_e = X_0 \ln 2. \tag{4}$$

The interaction of photons with matter is also simplified in this model. After traversing one splitting length d_e (the same as for electrons), a photon (whose energy must exceed E_e^{crit}) converts into an electron-positron pair, with the total photon energy divided equally between the two particles.

The development of an electromagnetic shower initiated by a single high-energy photon in matter is shown schematically in Fig. 1a. At the 0th step, there is one high-energy photon. After traversing one splitting length, the photon converts into an electron-positron pair. At the next step, after traversing another splitting length, the electron and positron each emit one photon, which in turn initiate their own showers. Thus, the shower continues to grow until the energies of the particles fall below the critical energy, at which point the development of the electromagnetic shower terminates.



(a) Electromagnetic shower (b) Hadronic shower

Fig. 1. Schematic illustration of shower development in air.

Problem

A photon with energy $E_\gamma = 10\ \text{GeV} = 10 \cdot 10^9\ \text{eV}$ enters air of constant density $\rho_0 = 1.3\ \text{kg/m}^3$. The electron charge is $q_e = -1.6 \cdot 10^{-19}\ \text{C}$.

A.0	Using the given value of the radiation length of air X_0, calculate the splitting length d_e. Express your answer in g/cm^2 and round it to one decimal place.	0.2pt

Note: We will call the number of shower generations n the quantity , which denotes how many successive splittings have occurred since the beginning of the shower development.

A.1	Determine the number of shower generations n_{max} after which the energy of the particles in the shower becomes less than the critical energy.	0.4pt

A.2	Determine the geometric distance l_e traveled by the shower (in m) in air by the time the maximum number of particles is reached.	0.4pt

A.3	Determine the maximum number of particles $N_{\max}$ in the shower.	0.2pt

A.4	Determine the total charge Q_e (in C) carried by the electrons in the shower by the time its development ceases.	0.8pt

Part B. Hadronic shower (2.0 points).

Theory

When charged hadrons (protons p or pions $\pi^{\pm}$) pass through matter, the situation becomes somewhat more complicated. In addition to electromagnetic interactions, hadrons also undergo inelastic interactions with the nuclei of the medium via the **strong interaction**. An inelastic interaction means that new particles are produced in the interaction, carrying away part of the energy. For example, the interaction of a proton p with a nucleus N can be represented schematically as

$$p + N \to N + \text{hadrons}. \tag{5}$$

In this problem, we will use a simplified model of inelastic interactions. Let us assume that, in an inelastic interaction with matter, the initial charged hadron (a proton or a pion $\pi^{\pm}$) transforms into 9 charged pions $\pi^{\pm}$ and 6 neutral pions π^0, with the energy distributed equally among them in the direction of motion of the initial hadron. Analogously to the electromagnetic shower, such a splitting occurs every **hadronic splitting length** d_h. In air, the hadronic splitting length is $d_h = 57\ \mathrm{g/cm^2}$.

After the splitting, the neutral pions π^0 immediately decay into two photons γ, with the energy shared equally between them. The charged pions $\pi^{\pm}$ continue to propagate and undergo inelastic interactions with matter every splitting length d_h until their energy falls below the critical energy E_π^{crit}. For pions in air, $E_\pi^{\mathrm{crit}} = 20\ \mathrm{GeV}$.

Once the critical energy is reached, the shower development ceases, and the charged pions decay into muons $\mu^{\pm}$ and neutrinos ν_μ, which carry away energy without interacting with the medium:

$$\pi^{\pm} \to \mu^{\pm}\nu_\mu. \tag{6}$$

Thus, when charged hadrons pass through air, a hadronic shower develops. It consists of an electromagnetic component (neutral pions decay into photons and initiate electromagnetic shower) and a hadronic component (charged pions generate further hadronic showers). The development of a hadronic shower initiated by a single high-energy proton in matter is shown schematically in Fig. 1b. A proton enters the medium and, after it travels one hadronic splitting length d_h, transforms into five π^+, four π^-, and six π^0 mesons. Each neutral pion then decays into two photons γ, which initiate electromagnetic showers, while the charged pions continue propagating until, after traveling another hadronic splitting length d_h, they generate the next generation of the hadronic shower. In this process, a π^+ produces five π^+, four π^-, and six π^0 mesons (just like the proton), whereas a π^- produces four π^+, five π^-, and six π^0 mesons. This process continues as long as the particles have sufficient energy, that is, until their energies fall below the corresponding critical values.

Problem

In a collider experiment, a leak occurred, and a proton with energy $E_p = 7\ \mathrm{TeV} = 7 \cdot 10^{12}\ \mathrm{eV}$ entered the atmosphere filled with air of density $\rho_0 = 1.3\ \mathrm{kg/m^3}$.

B.1	Determine the maximum number of generations of the hadronic component of the shower $n_{\mathrm{hadr}}^{\mathrm{max}}$.	0.4pt

B.2	Determine the maximum number of generations of the entire shower $n_{\mathrm{EM}}^{\mathrm{max}}$, taking into account the electromagnetic component.	1.0pt

B.3	Determine the fraction of the shower energy carried away by muons and neutrinos β_{miss}. Round your answer to two decimal places.	0.6pt

Part C. Air shower (3.0 points).

A cosmic-ray proton of ultra-high energy $E = 100\ \mathrm{TeV}$ enters the Earth's atmosphere vertically from above and initiates a hadronic shower. Assume the atmosphere to be isothermal, with an exponential density profile:

$$\rho(h) = \rho_0 e^{-\frac{h}{H}}, \tag{7}$$

where h is the altitude (in meters) above the Earth's surface, $\rho_0 = 1.3\ \mathrm{kg/m^3}$ is the air density at sea level, and $H = 8\ \mathrm{km}$ is the atmospheric scale height.

C.1	Determine the altitude h_{stop} at which the shower development ceases. Express your answer in kilometers and round it to one decimal place.	1.3pt

C.2	Determine what fraction of the shower energy, β_{EM}, ends up in the electromagnetic component. Round your answer to two decimal places.	0.5pt

After the shower development ceases, its radius can be estimated as

$$R_{\mathrm{stop}} = \frac{X_0}{\rho} \cdot \frac{E_s}{E_e^{\mathrm{crit}}}, \tag{8}$$

where $E_s = 21\ \mathrm{MeV}$.

Note: The shower radius is defined as the mean value of R over the shower particles:

$$R = \frac{\sum\limits_i \sqrt{x_i^2 + y_i^2}}{N},$$

where x_i and y_i are the geometric Cartesian coordinates of the shower particles relative to its center.

C.3	Estimate the shower radius R_{stop} at the altitude h_{stop}. Express your answer in meters and round it to the nearest integer.	0.2pt

Assume that the shower radius R decreases inversely proportional to the air density as it propagates downward without energy losses:

$$R \propto \frac{1}{\rho}. \tag{9}$$

C.4	Estimate the area of the shower S at the Earth's surface. Express your answer in m^2 and keep 2 significant figures.	0.2pt

Assume that the particles propagate at the speed of light $c = 3 \cdot 10^8\ \mathrm{m/s}$, and that the thickness of the shower front is determined by the difference in path lengths of particles diverging at an angle $\theta \approx R_{\mathrm{stop}}/h_{\mathrm{stop}}$.

C.5	Estimate the characteristic signal duration Δt at a detector located on the Earth's surface at the center of the shower. Express your answer in ns and round it to one decimal place.	0.8pt

Part D. Data Analysis (3.0 points).

To detect particles in high-energy physics, various types of detectors are used. In this problem, we consider a scintillation detector as an example. When a particle passes through such a detector, it produces optical photons (i.e., visible light), which are then detected by photomultiplier tubes (PMTs). In this system, a particle generates a digital signal in the detector that is proportional to its energy. One can introduce a proportionality coefficient κ. Neglecting detector noise, a particle with energy E striking the scintillation detector is registered with an amplitude A:

$$A = \kappa E.$$

Problem

A photon γ with initial energy E_0 enters the Earth's atmosphere vertically downward, toward the center of the Earth, and initiates an electromagnetic shower. A highly granular scintillation detector is installed on the ground. This means that the detector consists of many cells, each of which is a system composed of a scintillator and a PMT. The arrival coordinate of a particle is determined by which detector cell was triggered, while the signal amplitude is obtained from the PMT.

You are given a table with digitized detector data. Each row contains three variables:

A — signal amplitude,

x_{cm} — coordinate of the triggered detector along the x-axis (in cm),

y_{cm} — coordinate of the triggered detector along the y-axis (in cm).

Assume the atmosphere to be isothermal, with an exponential density profile as a function of altitude:

$$\rho(h) = \rho_{\mathrm{atm}} e^{-\frac{h}{H}}, \tag{10}$$

where ρ_{atm} is the air density at the Earth's surface and H is the atmospheric scale height.

D.1	Determine the coefficient κ of the detector at the Earth's surface.	1.0pt

D.2	Determine the energy of the initial photon E_0.	0.5pt

D.3	Calculate the atmospheric density at the Earth's surface ρ_{atm}.	1.5pt

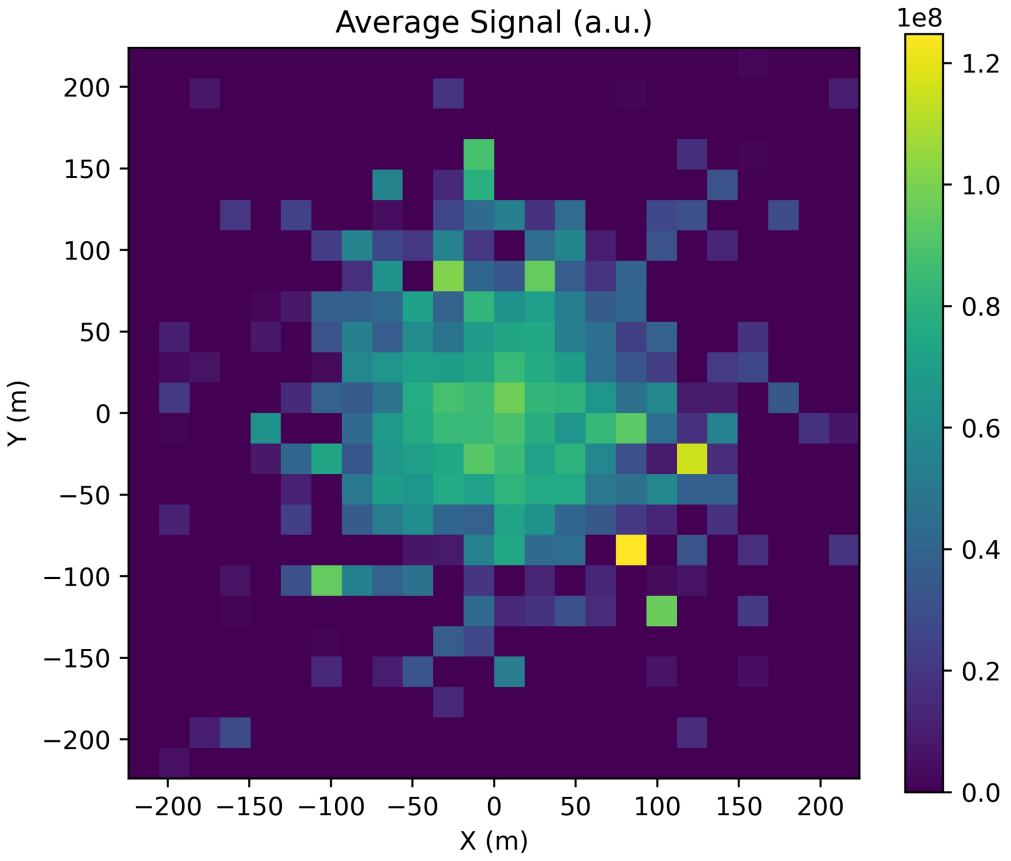


Fig. 2. Example of the 2D histogram of the detector signal at the Earth's surface.

PN junctions in semiconductors

Semiconductors are materials with electric conductivity significantly lower than conductivity of metals. Unlike insulators, conductivity of semiconductors is not zero and rapidly increases with temperature. This fact can be explained by the increase of charge carrier concentration due to thermal excitations.

There are two types of charge carriers in semiconductors, electrons and holes. Holes are positively charged particles, that appear in place where an electron is absent. Hole charge is equal to absolute value of electron charge. In a pure semiconductor concentrations of electrons and holes are equal to each other.

Semiconductor devices, in particular diodes and transistors, are widely used in modern electronics. In this problem we consider one of the simplest models of a semiconductor diode.

Charge carriers concentration in semiconductors can be easily modified by adding a small amount of impurities, i.e., atoms of another chemical element. Impurities that increase the electron concentration are referred to as donors, semiconductor with excess of donors is called n-type (from negative) semiconductor. Hole concentration in n-type semiconductor is much lower than electron concentration. Impurities that increase holes concentration are referred to as acceptors, semiconductor with excess of acceptors is called p-type (from positive) semiconductor.

Consider a contact of two semiconductors of p and n types (pn junction). In practice we can dope different parts of semiconductor with different types of impurities. The current through the junction depends on the polarity of applied voltage. The resulting structure is a semiconductor diode.

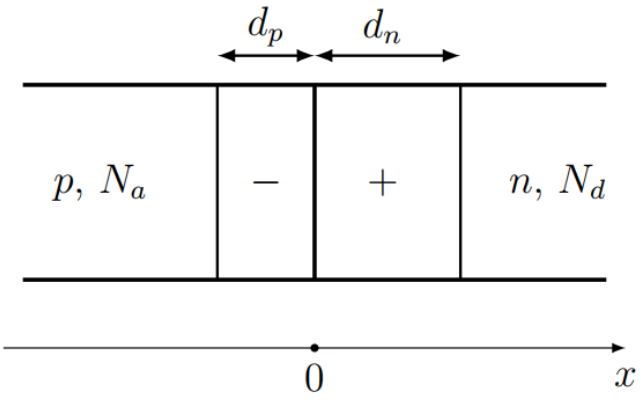


Fig. 1. Junction and depletion layers

In this problem we consider the following model of a diode (Fig. 1).

1. All physical quantities depend only on one coordinate x.
2. The region $x < 0$ is a p-type semiconductor with impurity concentration N_a.
3. The region $x > 0$ is a n-type semiconductor with impurity concentration N_d.
4. Near the junction electrons and holes diffuse through the boundary. As a result, in region $x < 0$ there is a negatively charged layer of width d_p, and in region $x > 0$ there is a positively charged layer of width d_n. These layers are called depletion layers.
5. In depletion layers the main charge carrier concentration (holes in p-type semiconductors and electrons in n-type semiconductor) is almost zero, outside of the layers their concentration is constant and equal to the concentration of impurities.

6. Volume charge density in depletion layer can be calculated from assumption that each donor impurity has charge $+q_e$, and each acceptor impurity has charge $-q_e$, where q_e – absolute value of electron charge.

You may use the following physical constants

- $\varepsilon_0 = 8.85 \cdot 10^{-12}\ \mathrm{F/m}$ – electric constant;
- $q_e = 1.60 \cdot 10^{-19}\ \mathrm{C}$ – absolute value of electron charge;
- $k = 1.38 \cdot 10^{-23}\ \mathrm{J/K}$ – Boltzmann constant;

For the silicon based diode, considered in this problem:

- $\varepsilon = 11.7$ – dielectric constant;
- $T = 300\ \mathrm{K}$ – temperature;
- $N_a = 1.0 \cdot 10^{23}\ \mathrm{m^{-3}}$;
- $N_d = 5.0 \cdot 10^{22}\ \mathrm{m^{-3}}$;
- $n_i = 1.0 \cdot 10^{16}\ \mathrm{m^{-3}}$ – concentration of electrons and holes in semiconductor without impurities.

Part A. Electrostatics (5.0 points)

In this part we consider electric field distribution in the junction without external voltage. From symmetry considerations, the electric field is parallel to the x axis, the only non-zero component is E_x.

A.1 Total electric charge near the junction must be equal to zero. Using this fact, write an equation connecting widths of depletion layers d_n, d_p and impurities concentrations N_a, N_d. 0.5pt

A.2 Show that the following equation holds for electric field 0.5pt

$$\frac{dE_x}{dx} = \frac{\rho}{\varepsilon_0 \varepsilon},$$

where ρ – volume charge density in the corresponding point.

A.3 Find the electric field $E_x(x)$ as a function of x. Assume that far from junction the field is equal to 0. Express the answer in terms of N_a, N_d, d_n, d_p, ε, ε_0 and q_e. 1.0pt

A.4 Find the electrostatic potential $\varphi(x)$ as a function of x. Assume that $\varphi(0) = 0$. Express the answer in terms of N_a, N_d, d_n, d_p, ε, ε_0 and q_e. Determine the potential difference at large distance to the junction $\Delta\varphi = \varphi(x = +\infty) - \varphi(x = -\infty)$. 1.0pt

A.5 Draw a qualitative plot of the potential $\varphi(x)$ as function of x. 0.5pt

A.6 Actually, the potential difference is determined by the material properties and the temperature. Knowing the potential difference, one can calculate depletion layers widths. Express the widths d_n, d_p in terms of $\Delta\varphi$, N_a, N_d, ε, ε_0 and q_e. Calculate their numerical values for $\Delta\varphi = 0.8\ \mathrm{V}$. 1.0pt

A.7 Determine the maximum value of the electric field in the junction $E_{\max}$. Express the answer in terms of $\Delta\varphi$, N_a, N_d, ε, ε_0 and q_e. Calculate its numerical value for $\Delta\varphi = 0.8$ V. 0.5pt

Part B. Junction with external voltage (5.0 points)

Now we apply voltage V to the junction (Fig. 2). The voltage is positive if applied to the p-type semiconductor. The applied voltage changes energies of electrons and holes in the p-type semiconductor, while energies in n-type semiconductor are the same. Assume that electrons and holes concentrations $n(x)$ and $p(x)$ obey the Boltzmann distribution:

$$n(x) = n_0 \exp\left(-\frac{U_n(x)}{kT}\right), \quad p(x) = p_0 \exp\left(-\frac{U_p(x)}{kT}\right),$$

where $U_n(x)$, $U_p(x)$ are potential energies of an electron or a hole in corresponding point, T – absolute temperature, n_0, p_0 – normalization constants, determined by the concentration of impurities N_d and N_a correspondingly. Concentration ratio of electrons and holes to the right and to the left of the junction is determined by energy difference:

$$\frac{n(x=+\infty)}{n(x=-\infty)} = \exp\left(-\frac{U_n(x=+\infty) - U_n(x=-\infty)}{kT}\right); \quad \frac{p(x=+\infty)}{p(x=-\infty)} = \exp\left(-\frac{U_p(x=+\infty) - U_p(x=-\infty)}{kT}\right).$$

If applied voltage $V > 0$, holes move form p-type semiconductor to n-type, while electrons move from n-type semiconductor to p-type. In addition, concentration of electrons in n-type semiconductor is determined by impurity concentration and does not change, while hole concentration increases. Similarly in p-type semiconductor hole concentration does not change while electron concentration increases.

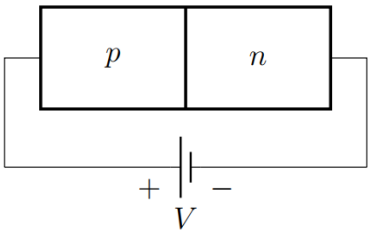


Fig. 2. Voltage applied to pn junction

B.1 Let n_{10} denote the concentration of electrons in p-type semiconductor without external voltage. Similarly let p_{10} denote the concentration of holes in n-type semiconductor without external voltage. Determine electron concentration n_1 in p-type semiconductor and hole concentration p_1 in n-type semiconductor after the voltage V is applied. Consider region outside of depletion layers, where electric field is zero. Express the answers in terms of p_{10}, n_{10}, V, T, k and q_e. 0.5pt

We obtained, that after the voltage is applied, the hole concentration in n-type semiconductor and electron concentration in p-type semiconductor are different from equilibrium ones. Electrons can move to

the vacant places, corresponding to holes. In such process the electron and the hole disappear (recombine). Boltzmann distribution does not take recombination into account. Therefore without diffusion charge carries concentrations change to equilibrium values.

To be specific, consider holes in n-type semiconductor outside of depletion layer. Denote the hole concentration by p_1, equilibrium concentration without diffusion by p_{10}. The rate of concentration change due to recombination can be expressed as

$$\left(\frac{dp_1}{dt}\right)_{\text{rec}} = -\frac{p_1 - p_{10}}{\tau_p},$$

where τ_p is time of hole recombination. Holes flux (number of holes that transfer through the unit area in unit time) due to diffusion is described by the formula

$$J_p = -D_p \frac{dp_1}{dx},$$

where D_p is holes diffusion coefficient.

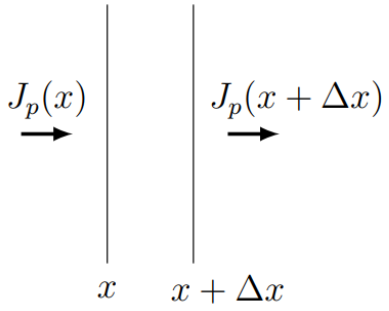


Fig. 3. Holes balance

B.2	Consider the volume between the planes with coordinates x and $x + \Delta x$ (Fig. 3). Write down the condition that the holes concentration in this volume does not change. Take into account the holes flux through the two planes and holes recombination. In the limit $\Delta x \to 0$ obtain a differential equation for the concentration deviation from equilibrium value $\Delta p = p_1 - p_{10}$ as a function of x. The answer may also include D_p and τ_p.	1.0pt

B.3	Find the distance L_p, over which the deviation of holes concentration from equilibrium decreases by e times. Express the answer in terms of D_p, τ_p. Compute numerical value for $D_p = 1.0 \cdot 10^{-3}\ \mathrm{m}^2/\mathrm{s}$, $\tau_p = 5.0 \cdot 10^{-7}\ \mathrm{s}$.	0.6pt

The characteristic length, over which the hole concentration changes due to recombination, is much larger than the depletion layer width. Therefore we can neglect recombination in depletion layer. The current through the junction can be calculated as a sum of electron current and hole current, independent of each other. To calculate the hole current we can consider n-type semiconductor near the depletion layer boundary, where the current is determined by diffusion and hole concentration can be calculated from Boltzmann distribution. The electron current can be found in the same way.

B.4	Determine the hole concentration $p_1(x)$ in region $x > d_n$ as a function of coordinate (in n-type semiconductor outside of the depletion layer). Express the answer in terms of p_{10}, V, T, L_p, d_n, k and q_e.	0.8pt

B.5	Determine the density of hole diffusion current j_p at the boundary of depletion layer $x = d_n$. Express the answer in terms of p_{10}, V, L_p, D_p, T, k and q_e.	0.8pt

The expression for the electron current can be obtained by analogy, if we replace hole concentration by electron concentration n_{10}, diffusion coefficient, recombination time and characteristic length by corresponding values for electrons, D_n, τ_n and L_n. Equilibrium holes and electrons concentrations are given by

$$p_{10} = \frac{n_i^2}{N_d}, \quad n_{10} = \frac{n_i^2}{N_a},$$

where n_i is concentration of electrons or holes in semiconductor without impurities.

B.6	The junction cross section area is A. Derive the expression for the total current I through the junction as a function of applied voltage V, semiconductor parameters N_a, N_d, τ_p, τ_n, D_p, D_n, n_i and constants k, q_e.	0.8pt

B.7	Draw a qualitative plot of the relation $I(V)$, obtained in previous task.	0.3pt

The numerical values of the parameters are

- $\tau_n = \tau_p = 5.0 \cdot 10^{-7}$ s;
- $D_n = 2.5 \cdot 10^{-3}$ m^2/s, $\quad D_p = 1.0 \cdot 10^{-3}$ m^2/s;
- $A = 1.0 \cdot 10^{-8}$ m^2.

B.8	Calculate the numerical values of the saturation current I_∞ at large negative voltage, and the value of current I_1 for $V = 0.7$ V.	0.2pt